

Novelty verdict: the multiples-of-four Liouville–Goldbach proof

Prepared 2026-09-28 for the accepted result

`ArithmeticStatement.representation_multiple_four` (every positive multiple of four is a sum of two positive integers with Liouville value -1), as proved in [PROOF.md](#).

Summary

Question	Verdict
Did the project depend on a library or repository that already contained this result?	No. No dependency found.
Is the theorem new?	No. CaptainSude’s all-even theorem (public 2026-09-16) implies it.
Is the proof strategy new?	No. The rigidity endgame (antireflection $\Rightarrow$ quadratic character $\Rightarrow$ contradiction at a small prime quadratic residue) is Mangerel’s, IMRN 2024.
Is any step new?	Plausibly yes. The bridge from a <i>missing Goldbach pair at $4p$ to full antireflection</i> (§§3–4) was not found in any source checked. The combinatorial multiplicativity argument (§5) is a new <i>method</i> for a known conclusion.

The contribution is best described as an independent, fully elementary proof of the multiples-of-four case, with an apparently new bridge from the Goldbach-type hypothesis to Mangerel’s rigidity setting. §2 must be credited to Mangerel (arXiv:2412.17199, §6.2) and §6 to Chowla–Cowles–Cowles (1980).

1. Dependency audit (no leakage found)

Evidence.

1. **No reference to the competing proof.** Outside the Lean build cache, a search of the problem directory for `CaptainSude` and `Astra` matches nothing except this verdict. The matches for `liouville-goldbach` are only the project’s own directory name `liouville-goldbach-jul2026`. No file imports or cites CaptainSude’s repository.
2. **Dependencies.** `lean/lakefile.toml` requires only Mathlib at `905b95818eb32af7874a58b427f50c1711a5e96c`. `lake-manifest.json` adds only Mathlib’s standard companions: `plausible`, `LeanSearchClient`, `importGraph`, `ProofWidgets4`, `aesop`, `Qq`, `batteries` and `Cli`.
3. **Mathlib contains no Goldbach-type Liouville result.** In the checked-out pin, `grep -ril goldbach Mathlib` returns only `NumberTheory/Fermat.lean`. Its “Goldbach’s theorem” is the unrelated fact that distinct Fermat numbers are coprime. `ArithmeticFunction.liouville` exists but is not used, because the project defines its own `lambda` from `primeFactorsList` in `Statement/Definitions.lean`.
4. **Timing.** The Mathlib pin was published 2026-07-28 (release workflow 30379912464), seven weeks before CaptainSude’s 2026-09-16 release. CaptainSude builds on a later Mathlib (`de2ef682...`) and Lean v4.34.0-rc2. This project uses Lean v4.32.2.
5. **Mathlib results actually used** are general and long-standing: quadratic reciprocity and the supplement at 2, `ZMod.exists_sq_eq_neg_one_iff`, `Nat.Prime.sq_add_sq`, `ZMod` field structure, finite pigeonhole, and factorization lemmas. See [sources.md](#) §§3–4 for line-level locators.
6. **Search hygiene.** `nl/searcher/search-trace.md` records one post-cutoff MathOverflow hit (revision 2026.9.15) that was excluded and not opened.

Limits. The provenance ledger was written by the agent team itself. This audit cannot inspect the base model’s training data.

Correction to an earlier informal claim. The *written* proof of §6 avoids quadratic reciprocity (it uses a Gauss-lemma floor count). The *Lean* proof does not: `Character/ResiduePrime.lean` uses Mathlib’s quadratic reciprocity, and the $p \equiv 1 \pmod{4}$ case uses `Nat.Prime.sq_add_sq`. The reciprocity-free argument is not formalized.

2. Sources compared

Source	Date	Read by the team before this audit?
A. P. Mangerel, <i>On a Goldbach-type problem for the Liouville function</i> , arXiv:2404.12117v2	2024-05-02	Yes, pages 1–4 and 14 (nl/searcher/extraction-v2.md)
A. P. Mangerel, same title, <i>IMRN</i> 2024(16):11865–11877, doi:10.1093/imrn/rnae149 (journal version, rewritten to be fully elementary)	2024-07-03	No
A. P. Mangerel, <i>On Shusterman’s Goldbach-type problem for sign patterns of the Liouville function</i> , arXiv:2412.17199v1	2024-12-23	No
S. Chowla, J. Cowles, M. Cowles, <i>The least prime quadratic residue and the class number</i> , J. Number Theory (cited as [2] in IMRN)	1980	No
CaptainSude, <i>Liouville-Goldbach</i> (GitHub, v1.0.0), attributed by the poster to “GPT-6 Astra”	2026-09-15/16	No (after the 2026-07-31 cutoff)

3. Step-by-step comparison

Proof step (PROOF.md)	Prior occurrence	Verdict
§2 reduction: $2m + 2m$ when $\lambda(m) = 1$; $8 = 3 + 5$ $= 4 + 4$ scaled for even m ; scale a $4p$ pair by d with $\lambda(d) = 1$	arXiv:2412.17199 §6.2, case (ii): “as $8 = 4 + 4 = 3 + 5 \dots a = b = 4N' \dots a = 3N', b = 5N'$ ”; case (iii): $a = b = N/2$	Known , except for the reduction to $4p$ for odd primes p
§3 rules (A), (B), (C), obtained by scaling pairs by 4 and by 2	IMRN Lemma 2.2 derives similar local identities by splitting $p = (p+m)/2 + (p-m)/2$ and $p = (2p+m)/3 + (p-m)/3$, but <i>from</i> the constancy of $\lambda(m)\lambda(p-m)$	Similar tools, different direction
§4 ternary gap descent $(x, y) \rightarrow ((p+x)/3, (p+y)/3)$, giving full antireflection $f(p-n) = -f(n)$ from a missing $-1/-1$ pair at $4p$	Not found. In IMRN, antireflection ($\lambda(m)\lambda(p-m)$ constant, eq. (3)) is the hypothesis $ \lambda(p) = p - 1$, not a consequence. IMRN Remark 2 says the methods “appear to be far too rigid” for the Goldbach-type problem	Plausibly new. This is the bridge that the prior work lacked
§5 antireflection $\Rightarrow$ multiplicativity on $(\mathbb{Z}/p\mathbb{Z})^\times$, by least bad multiplier and circular gaps ($nd < p$)	Same conclusion as IMRN Lemma 2.3, Proposition 2.4 and Lemma 2.5, but reached with Fourier coefficients, Plancherel, an iterative j -descent (§3), a primitive root and Gauss sums	Known conclusion, new elementary method. Possible folklore not ruled out
§6 a prime r dividing $(p+1)/4$ is a nonzero square mod p ($p \equiv 3 \pmod{4}$, $p \geq 7$)	IMRN Proposition 2.6 cites Chowla–Cowles–Cowles for “a prime $q \leq (p+1)/4 < p$ such that $\chi_p(q) = +1$ ”. CaptainSude chooses the same prime and proves it by reciprocity	Known. The Gauss-lemma proof is a self-contained re-proof
§7 contradictions: $F(-1) = -1$ against -1 being a square ($p \equiv 1 \pmod{4}$); $\lambda(r) = -1$ against r being a square ($p \equiv 3 \pmod{4}$)	IMRN Proposition 2.6: “for each prime $q < p$ we have $\chi_p(q) = -1$ ”, contradicted by a small prime residue	Known

4. Relation to CaptainSude’s proof

CaptainSude proves the stronger prime lemma “ $2p$ has a $+1/+1$ pair for every prime $p > 3$ ”. Doubling that pair yields this project’s $4p$ statement. Their argument also follows IMRN’s structure: first make multiplication by 2 and 3 exact (their “commuting square”, compare IMRN Lemma 2.3), then extend to all multipliers (their short-representative lemma, compare IMRN Proposition 2.4), and finish with the same $(p+1)/4$ prime. Their credit to “Mangerel’s rigidity and commuting-defect arguments” is accurate.

The two proofs are independent and differ in the middle step. This project’s §4 reaches **full** antireflection on $(0, p)$, which the $4p$ hypothesis permits because scaling by 4 preserves signs. CaptainSude works only on $(0, p/2)$ with an odd completion, because the weaker $2p$ hypothesis flips signs under scaling by 2.

5. Recommended attribution

- §2: Mangerel, arXiv:2412.17199, §6.2.
- §§5–7 overall strategy: Mangerel, IMRN 2024, §2 (rigidity via comparison with the Legendre symbol).
- §6: Chowla, Cowles and Cowles (1980), as cited in IMRN [2].
- Independence note: the team read only arXiv:2404.12117v2 (pages 1–4, 14). It did not read the IMRN version, 2412.17199 or Chowla–Cowles–Cowles, and nothing after the 2026-07-31 cutoff.

6. Open items

1. Search for folklore versions of the §5 argument (“multiplicative on short products implies a character”) before claiming novelty of method.
2. Any novelty claim for §4 should be checked by a specialist, for example Mangerel.
3. Formalize the reciprocity-free proof of §6 if the “no reciprocity” claim is to be made for the Lean artifact.